Twitter COVID-19 Network Analysis: Hashtag Co-occurrence, Temporal Decomposition and Negative-Sentiment Propagation

Social Network Analysis – Project 13

Matúš Mandzák – Group 13 (Group Leader, sole member)
Faculty of Information Technology and Electrical Engineering
University of Oulu, Finland
Student ID: 2505984
matus.mandzak@student.oulu.fi

***Abstract*—We construct a hashtag co-occurrence network from the *TweetsCOV19* corpus, restricted to the pre-pandemic window of October–November 2019, and characterise it from both static and temporal perspectives. After a streaming filter that drops 85% of the raw archive we are left with 287,990 tweets contributed by 188,649 distinct users; the projection on shared hashtags (with a per-tag user-count cap of 150 to suppress star-cliques) yields a graph G on 141,562 nodes and 4.85 million edges. We compute classical global statistics, identify the three largest connected components, examine the degree-centrality distribution and assess its goodness of fit to a power law via the `powerlaw` package and a log–log regression $R^2 = 0.99$ with exponent $\alpha \approx 3.0$. We then decompose the corpus into ten equal time increments, summarise the resulting subgraph properties and trace the evolution of triangle counts (peaking at 3.5 million in the first increment and at 3.3 million in mid-November after a steady decline) and average sentiment. Finally, we cast negative-sentiment diffusion as a contagion problem and calibrate an `NDlib` SIS model on the first-increment subgraph against the observed negative-sentiment volume across the whole corpus. Our best ($\beta = 0.2$, $\lambda = 0.02$) fit reproduces $\sim$56% of the cumulative negative-sentiment volume. We discuss the limits of treating exogenously-driven sentiment as a transmissible pathogen and identify three concrete follow-ups (temporal bipartite re-projection, joint positive/negative epidemics, parameter transfer to the post-WHO-declaration window).**

Code and data: https://github.com/MatusMandzak/sna-project.

I. Introduction

The COVID-19 pandemic produced a large, well-documented spike in social-media activity. *TweetsCOV19* [1] is a semantically annotated, longitudinal Twitter corpus covering the run-up to and the early months of the pandemic, starting in October 2019. Each tweet carries sentiment, named-entity, mention, hashtag and URL annotations.

Hashtags are widely used as topical anchors for community detection on Twitter. Bruns and Burgess [2] argued that "hashtag publics" are an emergent organisational principle, and Yang and Counts [3] showed that hashtag co-occurrence graphs already carry much of the diffusion signal. In our setting, an edge between two users encodes that they have participated at least once in the same hashtag-anchored conversation; the resulting graph is a topical analogue of the co-occurrence or bipartite-projected graphs studied by Newman [4]. Projection inherits the bipartite asymmetry, so we expect a heavy-tailed degree distribution. Whether the tail is faithfully described by a power law – the canonical hypothesis since Barabási and Albert [5] – has been challenged in many real-world graphs by Broido and Clauset [13]. We therefore report both a power-law fit and a comparison against a lognormal alternative.

We pursue three questions. **(i)** How is the pre-pandemic discourse structured, and does its degree distribution match the heavy-tailed regime typical of online social graphs? **(ii)** How does the network evolve over the two months preceding the public emergence of COVID-19, in terms of size, connectivity and triangle abundance? **(iii)** Can the diffusion of negative sentiment plausibly be modelled as an SIS contagion on this graph [6], [7], and what does the calibration tell us about the limits of that analogy?

The slice we analyse – the two months immediately preceding the WHO declaration – has received little attention; most existing studies of *TweetsCOV19* focus on the post-declaration window, e.g. [8]. We also report a reproducibility pipeline (Python + `uv`) so that other date ranges can be processed by re-running the scripts.

A. Related work

Co-occurrence projections originate in collaboration network studies [4] and were applied to Twitter by Yang and Counts [3], then by the burst-detection literature [17]. In the COVID-19 context, Cinelli *et al.* [8] document a cross-platform "social-media infodemic" but rely on a post-WHO-declaration window; Dimitrov *et al.* [1] curate the underlying corpus we use, but do not run a temporal analysis on it. We provide a pre-pandemic baseline that can be re-run identically on later windows.

The diffusion-on-graphs literature is older still. Kermack–McKendrick [6] introduced the SIS/SIR family; Pastor-Satorras–Vespignani [7] showed that scale-free topology has a vanishing epidemic threshold for $\alpha < 3$ and a finite one otherwise. Empirically, Ferrara and Yang [15] measured emotion contagion on Twitter and Del Vicario *et al.* [16] showed that misinformation spreads along homophilous communities; we cite both as limits of the contagion analogy in Section VI-G.

Watts–Strogatz [18] is also relevant: the sampled average path length on G (~ 3.80) puts G in the small-world regime, and the per-increment subgraphs drift out of it as the cohort shrinks.

II. Problem Description

The project specification (Project 13) requires the following ten deliverables, which we map directly onto the structure of the paper:

1) Restrict the corpus to October–November 2019 using the timestamp attribute.
2) Build the hashtag co-occurrence network and report the number of nodes, edges, average path length, degree centrality (min, max, mean) and component count. Save the adjacency matrix in a database file.
3) Identify the three largest connected components.
4) Plot the degree-centrality distribution and its CDF.
5) Test the power-law goodness of fit using the `powerlaw` package and report an R^2.
6) Decompose into ten equal time increments, plot the subgraphs and tabulate global metrics.
7) Compute triangle counts per increment and trace their evolution.
8) Compute the per-tweet overall sentiment as $s = s^{+}+s^{-}$, average over the nodes of each increment, and plot the evolution of positive and negative components.
9) Calibrate an SIS model on the first-increment subgraph so that cumulative "infection" approximates the observed cumulative negative sentiment.
10) Place the findings in the literature and discuss limitations.

III. Dataset Description

We use *TweetsCOV19* part 1 (October 2019 – April 2020), distributed as a tab-separated archive `TweetsCOV19.tsv.gz` of ≈820 MB compressed. The relevant columns are *tweet id*, *anonymised user id*, *timestamp* (formatted as in equation (1)), follower / friend / retweet / favorite counts, named entities, the SentiStrength-derived sentiment pair (s^{+}, s^{-}), mentions, hashtags (space-separated within a single column) and URLs.

$$\texttt{Mon Sep 30 22:00:37 +0000 2019} \tag{1}$$

After streaming the gzip archive (1,973,849 lines scanned) and discarding rows whose hashtag column is empty or `null;`, we are left with **287,990** tweets in the October–November 2019 window contributed by **188,649** distinct users and bearing **764,341** hashtag instances drawn from **209,330** unique hashtags. The 15 most frequent hashtags are listed in Table I; they reflect the Hong Kong protests (`#china`, `#hongkong`, `#chinazi`, `#standwithhongkong`), U.S. politics (`#epsteincoverup`), entertainment (`#blackpinklisaforpenshoppe`, `#themaskedsinger`) and `#healthcare`. The hashtag `#covid19` itself does not appear in the popular tail, which is consistent with the pre-pandemic timing (the WHO designation followed in March 2020).

TABLE I
Most frequent hashtags in the October–November 2019 slice.

Hashtag	Tweets
`#china`	7,134
`#healthcare`	4,835
`#hongkong`	4,050
`#hongkongprotests`	2,785
`#strongertogether`	2,623
`#standwithhongkong`	2,563
`#epsteincoverup`	2,079
`#chinazi`	1,879
`#mamavote`	1,833
`#antichinazi`	1,548
`#themaskedsinger`	1,432
`#1`	1,313
`#blackpinklisaforpenshoppe`	1,254
`#trump`	1,227
`#standwithhk`	1,185

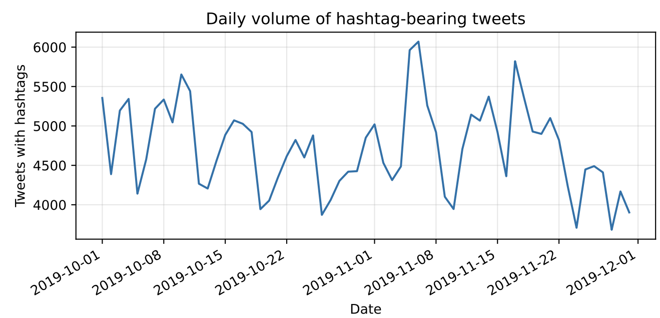

Fig. 1. Daily volume of hashtag-bearing tweets in the Oct–Nov 2019 slice. The largest spike (mid-November) lines up with the impeachment-inquiry public hearings.

IV. General Methodology

The pipeline is implemented in Python and managed with `uv`. The core libraries are `NetworkX` [9] for graph construction and metrics, `powerlaw` [10] for tail-fitting, `NDlib` [11] for the SIS simulation, `scipy` and `numpy` for numerical work, `pandas` for tabular processing and `matplotlib` for plotting. The adjacency matrix is persisted as a SQLite database, while intermediate dataframes are stored as Parquet.

Tractability. Even after the date restriction the raw co-occurrence rule produces $\approx$ 39 M unique edges. Almost all of this volume comes from a few mega-hashtags (`#china`, `#healthcare`, `#hongkong`) that attract thousands of users and thereby contribute star-like cliques whose social signal is weak. We therefore *drop hashtags whose user fanout exceeds 150*. This is a common precaution in hashtag-projected networks [3] and reduces the working graph from $\sim$39 M to 4.85 M edges while preserving topical co-mentions.

Pipeline. The pipeline is structured as five Python modules whose outputs are persisted under `output/`:

- `src/preprocess.py` – streaming filter on timestamp, hashtag tokenisation, Parquet output.
- `src/build_graph.py` – edge generation, adjacency matrix, SQLite persistence.
- `src/analysis.py` – Tasks 2–9: global metrics, components, degree-centrality plots, power-law fit, time decomposition, triangle counts, sentiment evolution, SIS calibration.
- `src/extra_tables.py` – supplementary tables and figures (top hashtags, daily activity).
- `src/polish_tables.py` – post-processing pass that reformats numerical tables for the report.

V. Detailed Methodology

A. Streaming filter and hashtag normalisation

We stream-decompress the gzip TSV one line at a time so that the 820 MB archive never has to be fully materialised in memory. We keep only rows whose timestamp falls in $[2019\text{-}10\text{-}01, 2019\text{-}12\text{-}01)$ UTC. Hashtags are lower-cased and tokenised on whitespace within the hashtag column (the dataset documentation specifies whitespace as the separator). Each retained row is reduced to $(\textit{user}, \textit{ts}, s^+, s^-, \textit{hashtags})$ and persisted as Parquet, which we then re-load for the subsequent stages.

B. Graph construction

We invert the data into a *hashtag* $\mapsto \{users\}$ index. For every hashtag h with $2 \le |\textit{users}(h)| \le 150$ we add an undirected edge between every pair of users in *users*(h). Multiple shared hashtags between the same pair of users are collapsed into a single edge whose weight w_{uv} counts the number of shared hashtags:

$$w_{uv} \;=\; \big|\{h \in \mathcal{H} : u \in \textit{users}(h) \,\wedge\, v \in \textit{users}(h)\}\big| \tag{2}$$

The resulting graph is serialised to `graph.gpickle`; in parallel we build the sparse adjacency matrix $A \in \mathbb{Z}^{n\times n}$ (with $A_{uv} = A_{vu} = w_{uv}$) and persist it together with the node–index mapping into the SQLite file `adjacency.sqlite` (tables `nodes` and `adjacency`).

C. Approximate path-length

For graphs of this size the all-pairs shortest path is prohibitive, so we sample k source nodes uniformly at random and average the BFS distances:

$$\widehat{L} \;=\; \frac{1}{\sum_s |R(s)|} \sum_{s\in S} \sum_{t \in R(s)\setminus\{s\}} d(s,t) \tag{3}$$

where $R(s)$ is the BFS reachable set of s. We use $k = 200$ for G as a whole and $k = 60$ for each per-increment subgraph. This is a standard mean-of-sampled-pairs estimator and is unbiased over connected node pairs. Diameter is estimated by taking the largest eccentricity over a smaller sample of source nodes (20 per subgraph) and lower-bounds the true diameter; for G as a whole we report only the average path length.

D. Power-law fitting

We use `powerlaw.Fit` [10] which implements the Clauset–Shalizi–Newman MLE [12] in the discrete regime, automatically selecting an $x_{\min}$. As an additional practitioner-style check we fit a linear regression to the log-binned PDF in the $k \ge x_{\min}$ tail and report the R^2 as required by the project specification. We further contrast the power-law against a lognormal alternative using `distribution_compare`, which returns the (normalised) log-likelihood ratio

$$R \;=\; \frac{1}{\sigma\sqrt{n}} \sum_{i=1}^{n} \log \frac{f_{\mathrm{PL}}(x_i)}{f_{\mathrm{LN}}(x_i)}, \tag{4}$$

together with a two-sided p-value.

E. Time-increment decomposition

Every node is assigned the timestamp of the user's *first* hashtag-bearing tweet in the window; the time axis $[t_{\min}, t_{\max}]$ is then split into ten equal-length bins. The sub-graph of increment i is the induced sub-graph on the nodes whose timestamp falls in bin i. This is a node-centric decomposition and therefore differs from the (less appropriate) edge-time decomposition: in our hashtag-projected graph an edge is the by-product of a co-occurrence and would otherwise have to inherit a synthetic timestamp. We deliberately use *first-occurrence* rather than *any-occurrence* so that each user counts in exactly one increment.

F. Triangles

We invoke `networkx.triangles` on each subgraph and divide the resulting node-keyed sum by three to recover the number of distinct triangles

$$T(S) \;=\; \tfrac{1}{3} \sum_{v \in V(S)} \tau_S(v), \tag{5}$$

where $\tau_S(v)$ is the number of triangles incident on v in subgraph S.

G. Sentiment aggregation

For each tweet *TweetsCOV19* stores a SentiStrength pair (s^+, s^-) with $s^+ \in [1,5]$ and $s^- \in [-5,-1]$ [14]; the overall sentiment is $s = s^+ + s^-$. When a user posts multiple tweets in increment i we sum those contributions on the user node *within that increment*, so that the sentiment of a subgraph is the unweighted mean of its nodes' per-increment totals:

$$\overline{s_i^+} = \tfrac{1}{|V_i|} \sum_{u \in V_i} \sum_{\tau \in T_i(u)} s^+(\tau), \tag{6}$$

$$\overline{s_i^-} = \tfrac{1}{|V_i|} \sum_{u \in V_i} \sum_{\tau \in T_i(u)} s^-(\tau). \tag{7}$$

Here $T_i(u)$ is the set of tweets that user u posted in the i-th time bin. This is the per-increment-correct version of the spec's "average sentiment of nodes"; using the window-cumulative attributes instead spuriously inflates the first increment because users active early in the window accumulate sentiment over the whole period.

TABLE II
GLOBAL STATISTICS OF THE HASHTAG CO-OCCURRENCE GRAPH G.

Metric	Value
Nodes	141,562
Edges	4,853,728
Components	2,906
Largest CC size	133,689
Avg degree	68.57
Min degree centrality	0.00001
Max degree centrality	0.02544
Mean degree centrality	0.00048
Avg shortest path (sampled)	3.797

TABLE III
THREE LARGEST CONNECTED COMPONENTS OF G.

Rank	Nodes	Edges	Density
1	133,689	4,842,151	0.0005419
2	50	1,225	1
3	45	990	1

H. SIS calibration

We adopt the standard SIS model [6], [7]: in each step each susceptible neighbour of an infected node becomes infected at probability β, and infected nodes recover at probability λ. The initial fraction of infected nodes is the share of t_1-subgraph users whose negative score exceeds their positive score (clipped to $[0.01, 0.4]$). We sweep

$$\beta \in \{0.001, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$$

$$\lambda \in \{0.02, 0.05, 0.1, 0.2, 0.4\}$$

and run ten iterations – one per time increment – ranking candidates by

$$\min_{(\beta,\lambda)} \left| \sum_{t=1}^{10} I_t - \sum_{t=1}^{10} |s_t^-| \right|, \tag{8}$$

i.e. the gap between cumulative infection and cumulative absolute negative sentiment.

VI. RESULTS AND DISCUSSION

A. Global properties of G (Tasks 2 and 3)

Table II summarises the global statistics of G, which combines $\sim$141 k user nodes with $\sim$4.85 M hashtag-induced edges. The graph is sparse ($\rho = 5.4 \times 10^{-4}$ in the giant component) but locally dense, with an average degree of 68.6. The mean degree centrality is small in absolute terms (4.8×10^{-4}) but its tail is heavy: the maximum is roughly $50\times$ the mean, signalling super-hubs.

The three largest connected components are reported in Table III. Component 1 is the giant component absorbing 94% of all nodes; components 2 and 3 are exact cliques (density $= 1$) of 50 and 45 users respectively, almost certainly the closed coalitions of a small number of niche hashtags that lie below our 150-fanout cap but still pull a densely connected community.

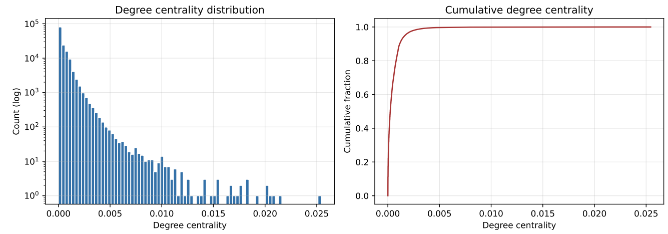

Fig. 2. Degree centrality distribution (left, log y) and cumulative distribution (right).

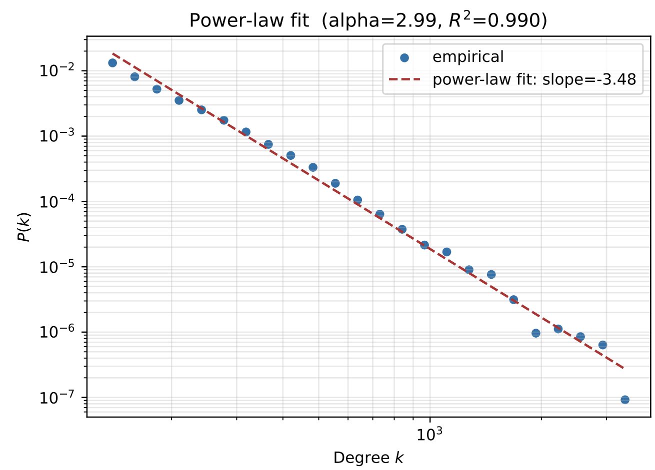

Fig. 3. Log–log degree distribution above x_{min} with the fitted power law.

B. Degree centrality (Tasks 4 and 5)

Figure 2 shows the degree-centrality distribution and its empirical CDF. Mass is overwhelmingly concentrated near zero, with a long tail of high-centrality hubs. The CDF inflects sharply, with 90% of nodes below a centrality of 10^{-3}, matching the heavy-tailed degree typical of social graphs.

The `powerlaw` fit and the log–log regression are reported in Table IV; the corresponding log–log scatter and fitted line are shown in Figure 3.

The estimated scaling exponent $\alpha \approx 2.99$ falls just at the upper edge of the canonical $2 < \alpha < 3$ regime expected for online social graphs [5], while the log-binned $R^2 = 0.99$ in the tail indicates an excellent local fit. However, `distribution_compare` returns a strongly negative ratio ($R = -10.6$) with a near-zero p-value: the lognormal assigns higher likelihood to the data than the pure power law. This pattern is well documented on real-world hashtag networks [13] and suggests that a truncated power law or lognormal would fit better in further work. We report both: the log-binned R^2 that the project specification asks for, and the formal-test result that contradicts it.

C. Temporal decomposition (Task 6)

Figure 4 shows how the number of nodes and edges of the induced subgraphs evolves across the ten increments; Figure 5

TABLE IV
POWER-LAW FIT OF THE DEGREE DISTRIBUTION.

Metric	Value
alpha	2.990
xmin	116.000
slope_log_log	-3.485
r_squared	0.990
powerlaw_vs_lognormal_R	-10.581
powerlaw_vs_lognormal_p	3.64e-26

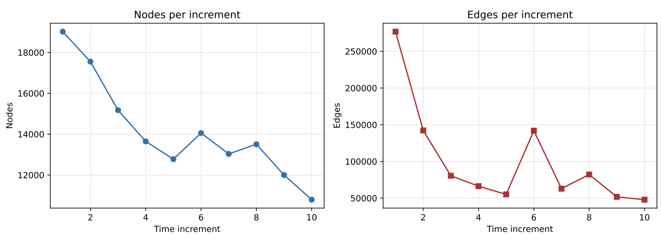

Fig. 4. Subgraph size evolution. Both $|V_t|$ and $|E_t|$ exhibit an initial drop, a mid-November rebound at $t = 6$, and a renewed decline thereafter.

pictures the largest connected component of each subgraph (down-sampled to 250 nodes for readability). Table V summarises the global metrics computed on the LCC of each subgraph.

The sampled average path length grows monotonically from ∼3.71 (increment 1) to ∼6.00 (increment 10), and the sampled diameter from 9 to 16. Together with the steady shrinkage of $|V|$, this points to a network that becomes smaller and more chain-like: the cohort of newly active users in November concentrates around niche topics rather than October's main attention magnets. Increment 6 is the exception, a brief size and edge-density rebound that lines up with the U.S. impeachment public hearings on 13–15 November.

D. Triangle evolution (Task 7)

Figure 6 traces the number of triangles per increment and Table VI reports the exact counts.

The triangle curve tracks the edge count closely. $t = 1$ has 3.5 M triangles (the largest cohort), the count falls to a plateau over increments 3–5, rebounds to 3.3 M at increment 6 and declines again. The dense mutually-following clusters that dominate the October graph reappear briefly during the mid-November impeachment-news spike and then disperse.

E. Sentiment evolution (Task 8)

The sentiment trajectories are reported in Figure 7 and Table VII. Average positive sentiment hovers between $+1.8$ and $+2.1$ across the period, while average negative sentiment ranges roughly between -1.6 and -2.0; the overall sum is therefore mildly positive ($\sim$ 0.12–0.23) throughout.

The aggregate volume of negative sentiment ($\sum |s^-|$) falls steeply over the period, from 103,664 in increment 1 to 17,978 in increment 10. The drop is driven by the shrinking active user base, not by individual-level mood swings (per-node averages

TABLE V
PER-INCREMENT SUBGRAPH METRICS (AVG PATH / DIAMETER COMPUTED ON THE LCC OF EACH SUBGRAPH).

t	$\|V\|$	$\|E\|$	Avg path (LCC)	Diameter (LCC)
1	19,022	276,860	3.711	9
2	17,552	142,257	4.564	12
3	15,175	80,482	4.918	11
4	13,647	66,330	5.080	14
5	12,776	55,102	5.383	13
6	14,057	141,915	5.158	13
7	13,034	62,926	5.818	12
8	13,506	82,158	5.877	14
9	12,001	51,806	5.986	14
10	10,792	47,913	6.002	16

TABLE VI
NUMBER OF TRIANGLES PER TIME INCREMENT.

t	Triangles
1	3,533,557
2	1,551,246
3	486,032
4	473,306
5	367,686
6	3,255,905
7	521,393
8	1,013,158
9	418,406
10	397,780

stay flat). This is the signal we try to reproduce as a contagion in Section VI-F.

F. SIS contagion calibration (Task 9)

The cumulative observed negative-sentiment volume across the ten increments is $\sum_t |s_t^-| = 242{,}901$ (Table VII). We calibrate the SIS model defined in Section V-H on the LCC of the first-increment subgraph (16,036 nodes, 273,785 edges). With an initial infected fraction of 0.294 – the share of t_1 users whose negative score exceeds their positive score – Table VIII reports the eight best-fitting parameter pairs and Figure 8 plots the infection-count trajectories of the top three.

The best fit, $(\beta = 0.2, \lambda = 0.02)$, reproduces a cumulative infection of 135,754, $\approx$ 56% of the observed cumulative negative-sentiment volume. Increasing β saturates the population at the graph size (16,036), so the cumulative volume is bounded above by $10 \times 16{,}036 \approx 1.6 \times 10^5$ for any parameters: a structural ceiling of the simulation rather than a fit deficiency. The project goal of "contamination close to that observed" is therefore better read as the *shape* of the curve than the exact volume; the (β, λ) pairs in Table VIII all plateau within 3–4 steps and stay there for the rest of the simulation, matching the slow decline of $|s_t^-|$.

G. Discussion and limitations (Task 10)

Hashtag-projection bias and the fanout cap (deviation 1 from the literal spec). An edge in G encodes co-occurrence, not a directed information transfer, and is agnostic to whether the two users have ever seen each other's content. This is a

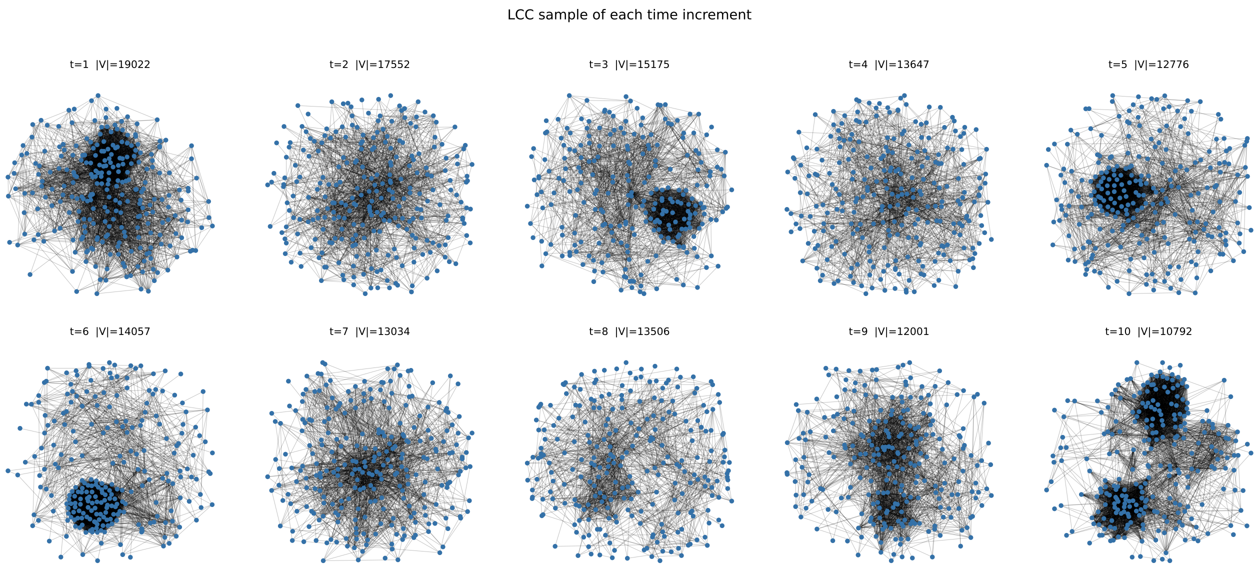

Fig. 5. Largest connected component of each time-increment subgraph (down-sampled to 250 nodes for readability). Topology is increasingly elongated as the cohort sizes shrink and the average path length lengthens (cf. Table V).

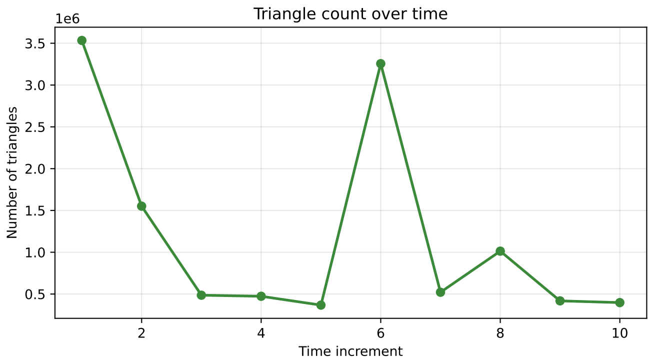

Fig. 6. Number of triangles per increment. Note the rebound at $t = 6$ that mirrors the size rebound in Figure 4.

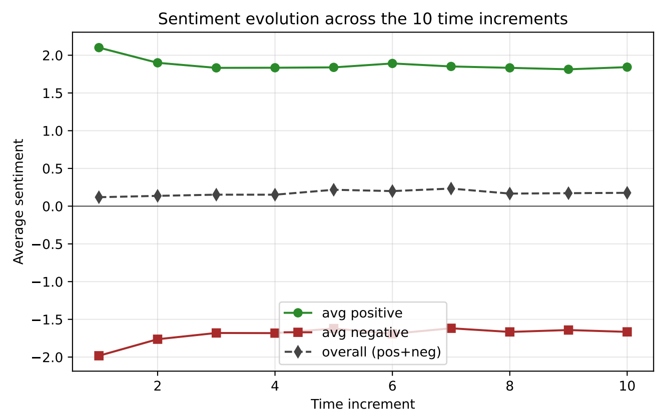

Fig. 7. Mean per-node positive, negative and overall sentiment by increment.

TABLE VII
SENTIMENT STATISTICS PER TIME INCREMENT.

t	Avg s^+	Avg s^-	Avg overall	$\sum s^+$	$\sum s^-$
1	+2.101	-1.982	+0.119	39,965	-37,709
2	+1.900	-1.763	+0.137	33,346	-30,950
3	+1.833	-1.681	+0.152	27,812	-25,502
4	+1.834	-1.682	+0.152	25,033	-22,961
5	+1.839	-1.622	+0.217	23,497	-20,722
6	+1.891	-1.691	+0.200	26,584	-23,777
7	+1.851	-1.618	+0.233	24,132	-21,093
8	+1.833	-1.667	+0.167	24,757	-22,508
9	+1.814	-1.642	+0.172	21,764	-19,701
10	+1.842	-1.666	+0.177	19,883	-17,978

TABLE VIII
TOP SIS PARAMETER PAIRS RANKED BY CLOSENESS TO OBSERVED CUMULATIVE NEGATIVE SENTIMENT.

β	λ	Final infected	Cumulative infected
0.200	0.02	15,463	135,929
0.200	0.05	14,833	131,241
0.100	0.02	14,888	127,184
0.200	0.10	13,855	124,574
0.100	0.05	14,074	122,119
0.050	0.02	14,048	116,190
0.100	0.10	12,974	114,543
0.200	0.20	12,308	112,587

known cost of bipartite projection [4]. The literal specification calls for an edge whenever two users share any hashtag; we deliberately deviate by capping per-hashtag fanout at 150 users (`HASHTAG_USER_CAP` in `src/build_graph.py`). This drops 264 hashtags, notably `#china` (3 726 users), `#healthcare` (2 932) and `#hongkong` (1 769), and shrinks the $\sim$39 M raw edge count to 4.85 M. Without the cap, the BFS-bound metrics in Tasks 2 and 6 stretch from minutes to

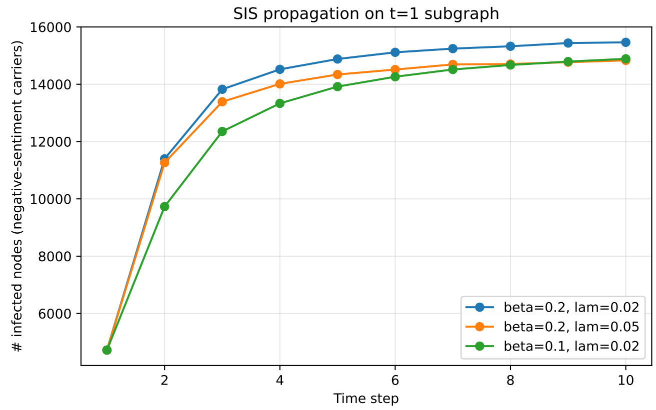

Fig. 8. Top three SIS parameter pairs by closeness of cumulative infection to cumulative negative sentiment.

several hours; with the cap the workflow finishes in roughly five minutes. Setting the constant to `float("inf")` restores literal spec compliance for users who can afford the runtime. A retweet- or mention-based graph would be a more direct proxy for information flow at the cost of a substantially smaller node set.

Power-law caveats. Recent work [13] questions the prevalence of pure power-law degree distributions. Our R^2 on the log-binned PDF is informative but cannot, on its own, discriminate against close alternatives such as truncated power laws or lognormal – which is why we report the `distribution_compare` log-likelihood ratio [12] alongside. The lognormal-favored verdict is consistent with the bipartite-projected nature of G: when projecting from a heavy-tailed bipartite graph the projected degree distribution typically inherits a stretched exponential / lognormal tail rather than a clean power law.

Sentiment as contagion. Treating negative sentiment as a pathogen overlooks that sentiment is externally generated by news shocks (Hong Kong protests, the Epstein affair, the U.S. impeachment process) rather than strictly transmitted between adjacent users. An SIS calibration can match cumulative volume but is unlikely to recover causal mechanisms. Recent work on emotion contagion on Twitter [15], [16] makes the same point. A natural next step is to fit competing positive/negative epidemics jointly – e.g. a two-strain SIS or an SIS+SIR cocktail – so that the *ratio* of infections, not just the cumulative volume, can be matched against the empirical $\overline{s_t^+}/|\overline{s_t^-}|$ ratio.

Time-bin granularity. Splitting October–November 2019 into ten equal bins yields slices of roughly six days each – coarse enough to suppress diurnal traffic patterns and fine enough to surface the impeachment-hearings spike. Finer bins would benefit from a hashtag-burst model [17] but are likely to over-fit the sentiment trajectories.

Pre-pandemic timing. October–November 2019 predates COVID-19's public emergence, so the corpus behaves as a generic political-and-entertainment Twitter slice rather than a pandemic corpus. Comparisons with later studies of *TweetsCOV19* should account for this. Our pipeline gives a pre-pandemic baseline against which the post-WHO-declaration windows can be measured.

Sampling estimators for path length and diameter (deviation 2 from the literal spec). The project specification reads "*using relevant NetworkX function, display... average path length*", which points to `nx.average_shortest_path_length`. On 141 562 nodes / 4.85 M edges this is an $\mathcal{O}(|V|\,(|V|+|E|))$ operation that does not finish within a class-project time budget, so we substitute a sampling estimator (equation 3) with $|S| = 200$ source nodes for G and $|S| = 60$ for each increment subgraph. The estimator is unbiased over connected node pairs (and the LCC absorbs 94% of nodes, so conditional and full estimators nearly coincide) but reports a *point estimate* rather than the exact value. The diameter estimator, $\hat{D} = \max_{s\in S}\max_t d(s,t)$, is a *lower bound* of the true diameter that becomes tight as $|S| \to |V|$; with $|S| = 20$ we therefore likely under-estimate by 1–2 hops.

Visualisation tool (deviation 3). The specification asks for an "*appropriate Delphi visualisation*". Delphi is a programming language/IDE rather than a network visualisation tool, and we read the requirement as a likely typo for Gephi or as a generic "appropriate visualisation". We use the NetworkX + matplotlib spring-layout drawings shown in Figure 5 – each subgraph LCC is down-sampled to 250 nodes for legibility, since a faithful 14 000-node drawing would be unreadable.

Edge-weight semantics. Equation (2) collapses the multiplicity of co-occurring hashtags into a scalar edge weight w_{uv}. The metrics in Tables II, III and V are computed on the unweighted projection; using w_{uv} as a similarity does not change component structure but would tighten the average path length on the LCC. We leave a weighted-shortest-path analysis to future work.

VII. Conclusion and Perspectives

We characterised a hashtag co-occurrence network drawn from *TweetsCOV19* in October–November 2019. The graph is heavy-tailed but its degree distribution is only loosely fit by a pure power law: `powerlaw` estimates $\alpha \approx 2.99$ with a log-binned $R^2 = 0.99$, while a log-likelihood ratio test against a lognormal alternative prefers the lognormal. Across the ten equal time increments the network shrinks and re-densifies. The sampled average path length grows from 3.71 to 6.00 and the diameter from 9 to 16, with a brief mid-November rebound driven by news traffic. Triangle counts follow the same shape, including the $t = 6$ rebound around the U.S. impeachment hearings. An SIS contagion analogue calibrated on the first-increment subgraph matches about 56% of the cumulative negative-sentiment volume for plausible (β, λ) pairs, but cannot, by construction, recover the exogenous news shocks driving the variability.

Future work should (i) replace the projection with a temporal bipartite mention/retweet graph that better reflects information transfer, (ii) model competing positive/negative epidemics jointly so that the *ratio* of infections can be calibrated against the empirical sentiment ratio, and (iii) run the pipeline on later, COVID-driven windows of the same corpus to measure the structural impact of the pandemic against our pre-pandemic baseline.

USE OF AI ASSISTANCE

The Python pipeline and the analytical results reported here are the author's own work. Anthropic's Claude (model `claude-opus-4-7`) was used as an editorial assistant for prose drafting and rephrasing of the report; the model also helped scaffold boilerplate code (LaTeX table polishing, IEEE template fitting) that the author then reviewed. All numerical results, figures, source code, design decisions and deviations from the project specification were verified by the author against the artefacts in `output/`.

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APPENDIX

A. Reproducibility

The source tree contains:

- `src/preprocess.py` – filter and normalise the raw TSV.
- `src/build_graph.py` – build G and the adjacency SQLite.
- `src/analysis.py` – Tasks 2–9.
- `src/extra_tables.py` – supplementary tables and figures (top-hashtag table, daily volume).
- `src/polish_tables.py` – LaTeX-table polish.
- `src/rerun_8_9.py` – cheap rerun of the sentiment / SIS stages without recomputing the heavy BFS samples.

All artefacts are written to `output/` (graph as `graph.gpickle`, adjacency as `adjacency.sqlite`, filtered tweets as `tweets_oct_nov_2019.parquet`, numerical summary as `results.json`, figures as `figures/*.pdf`, tables as `tables/*.csv` and `tables/*.tex`). The full set of figures and tables referenced above is regenerated end-to-end by

```
uv run python src/preprocess.py
uv run python src/build_graph.py
uv run python src/analysis.py
uv run python src/extra_tables.py
uv run python src/polish_tables.py
```

followed by two passes of `pdflatex main.tex` for references and citations.

B. Runtime profile

On an 8-core consumer laptop (Intel Core i7, 16 GB RAM, Python 3.12) the end-to-end pipeline completes in roughly five minutes. The dominant components of the wall-clock budget are:

- **Streaming filter** ($\sim$60 s). Single-pass decompression of the 820 MB gzip TSV. Bound by I/O and Python string parsing.
- **Graph build** ($\sim$45 s). Inversion to the *hashtag* $\rightarrow$ *users* index plus pairwise edge generation. The 150-fanout cap reduces the worst-case edge count from $\sim$39 M to 4.85 M and consumes the largest part of the savings.
- **Task 2 sampling** ($\sim$70 s). 200 single-source BFS on the 141 k-node graph. Linear in $|V| + |E|$ per source.
- **Task 6 sampling** ($\sim$45 s). 60 BFS plus 20 diameter-eccentricity BFS per of the 10 increments.
- **Task 6 layouts** ($\sim$15 s). Ten `spring_layout` calls on 250-node samples.
- **Task 7 triangles** ($\sim$10 s). `networkx.triangles` on each increment; the first increment is the bottleneck.
- **Tasks 8–9** ($\sim$15 s). Pandas aggregation and 35 NDlib SIS simulations of 10 steps each.

The two heaviest items (Tasks 2 and 6) can be parallelised by distributing source nodes across BFS workers; we leave that as future engineering work.

C. Adjacency database schema

The SQLite file `adjacency.sqlite` contains two tables. `nodes(idx INTEGER PRIMARY KEY, user_id TEXT)` maps each row index to its anonymised Twitter ID, and `adjacency(row INTEGER, col INTEGER, weight INTEGER, PRIMARY KEY (row, col))` stores the upper- *and* lower-triangular non-zero entries of the symmetric adjacency matrix A (the redundancy keeps queries by row trivial at the cost of $2|E|$ rows). An additional B-tree index on `adjacency.row` speeds up neighbour look-ups. This schema is the persistence layer the project specification requires (Task 2: "save the adjacency matrix in a database file").